The organisms that retain the colour of the primary stain are called Gram positive and those that do not retain the primary stain when decolorised and take on the colour of the counter stain are called Gram negative.

Mordants: Mordants are not dyes. They are important to increase the biological specimen's affinity for a dye. Some stains never stain the cells or its components unless treated with a mordant. The mordant becomes attached to a cell or its components and then combines with the stain to form an insoluble colour complex.

3.7.2 Procedure of Gram's Staining

Gram's Staining comprises of four steps:



Step 1: A heat fixed smear is covered with a basic violet dye, Example: Crystal violet. This stain imparts its colour to all cells. It is referred to as a primary stain, since it is applied first.

Step 2: After a short time, the slide is washed off and the smear is covered with iodine, a mordant. At this stage both Gram positive and Gram negative bacteria

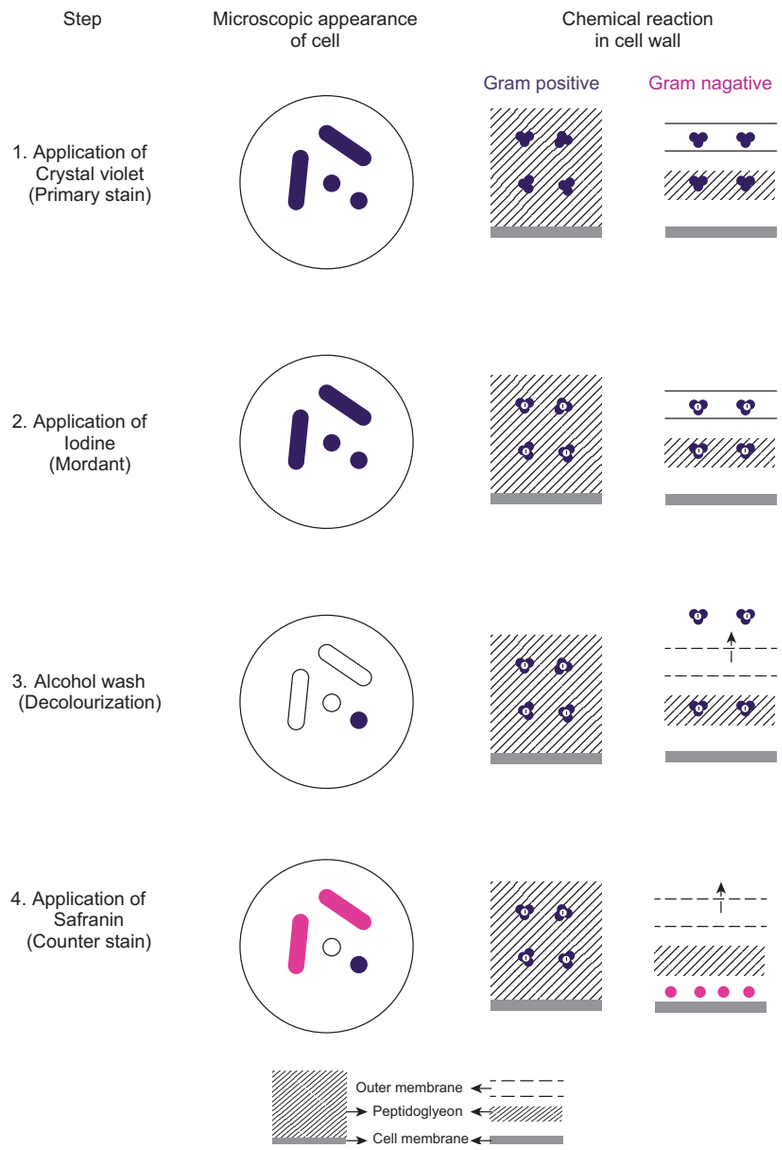


Figure 3.7: Steps, micrograph and chemical reaction of Gram Stained Bacteria

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